

Gibbs sampling: (Markov Chain Monte Carlo)

Want to simulate/generate random vectors from a distr. Π (a probability on $\mathbb{R}^k$)

We cannot invert cdf.

eg. $k=2$. $F(a,b) = F(c,d)$ for $(a,b) \neq (c,d)$

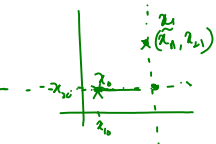
$U \sim \text{Unif}(0,1)$. Inverse does not exist.

Multivariate normal

Multinomial

$\Pi(x_1, \dots, x_k)$

$\tilde{x}_0 = (x_{10}, \dots, x_{k0})$



$p(x_0, y)$

For $x_{20}, \dots, x_{k0}$ $x_1 \sim \Pi_1(\cdot | x_{20}, \dots, x_{k0})$ univariate invert cdf.

$\tilde{x}_1 = (x_{11}, \dots, x_{k1})$ $x_2 \sim \Pi_2(\cdot | x_{11}, x_{30}, \dots, x_{k0})$

$\tilde{x}_2$ $x_{k1} \sim \Pi_k(\cdot | x_{11}, x_{21}, \dots, x_{k-1,1})$

$\tilde{x}_n, \dots, \tilde{x}_{n+m} \sim \Pi$

eg. $\Pi = \mathcal{N}_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$ $\begin{pmatrix} x \\ y \end{pmatrix} \sim \Pi$

$$-\frac{1}{2} (x \ y)^T \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}^{-1} \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2} y^T y \sqrt{\frac{1}{1-\rho^2} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}}$$

$x|y \sim \mathcal{N}(\rho y, 1-\rho^2)$

$y|x \sim \mathcal{N}(\rho x, 1-\rho^2)$

eg. $\tilde{x}_0 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \in \mathbb{R}^2$ First step $x_1 \sim \mathcal{N}(\rho y_0, \sqrt{1-\rho^2})$ $y_1 \sim \mathcal{N}(\rho x_1, \sqrt{1-\rho^2})$ $\tilde{x}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$

eg. $x_0 \sim \mathcal{N}(0,1) \parallel y_0 \sim \mathcal{N}(0,1)$ $x_2 \sim \mathcal{N}(\rho y_1, \sqrt{1-\rho^2})$ $y_2 \sim \mathcal{N}(\rho x_2, \sqrt{1-\rho^2})$ $\tilde{x}_2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$

eg2. 1-dimensional Ising model. $\tilde{x} = (x_1, \dots, x_d)$ d particles. Temp

Each particle has a characteristic $x_i \in \{+1, -1\}$ eg spin

$$\Pi(\tilde{x}) \propto \exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right)$$

How to generate configurations $\tilde{x}$ from the distr Π .

$$\Theta = E[R(\tilde{x})]$$

$$\Pi(\tilde{x}) = c \exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right) \frac{1}{2} (x_2 + x_3 + \dots + x_d)$$

$$\Pi_2(x_2^* | \tilde{x}_{[-2]}) = \frac{c \exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right)}{c \exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right) + c \exp\left(\frac{1}{T} \sum_{i=1}^{d-1} x_i x_{i+1}\right)}$$

$$\tilde{x}_{[-2]} = (x_1, x_3, \dots, x_d)$$

$$\Pi_2(x_2^* | \tilde{x}_{[-2]}) = \frac{\exp\left(\frac{1}{T} (x_1 + x_3)\right)}{\exp\left(\frac{1}{T} (x_1 + x_3)\right) + \exp\left(\frac{1}{T} (x_1 + x_3)\right)}$$

$$\Pi_1(x_1 = +1 | \tilde{x}_{[-1]}) = \frac{e^{\frac{1}{T} x_2}}{e^{\frac{1}{T} x_2} + e^{\frac{1}{T} x_2}}$$

Start with some configuration. Run the chain.

At n th step $\Pi_i(x_i = -1 | \tilde{x}_{[-i]}) = p_i$. Generate $U \sim U(0,1)$.

If $U < p_i$, then $x_i = -1$
else $x_i = +1$.